



Course Plan

Semester: IV

Year: **2025-26**

Course Title: Object Oriented Programming	Course Code: 24ECAC207
Total Contact Hours: 40	Duration of ESA: 3 Hours
ISA Marks: 50	ESA Marks: 50
Lesson Plan Author:Maheshgouda Patil	Date: 29/01/2026
Checked By: Dr. Narayan D.G.	Date: 29/01/2026

Prerequisites: Knowledge of

- i. C Programming
- ii. Data Structures and Algorithms

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Describe the object-oriented programming paradigm.
2. Apply object oriented principles to design a solution using JAVA.
3. Design reusable code using Java packages, interfaces and collection framework.
4. Apply functional programming principles to design serverless web services.
5. Design object-oriented solution using UML for real-world problems using system thinking.



Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Object Oriented Programming	Semester: IV
Course Code: 24ECAC207	Year: 2025-26

Course Outcomes (COs) / Program Outcomes (POs)	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Describe the object-oriented programming paradigm.	M												
2. Apply object oriented principles to design a solution using JAVA.		H											
3. Design reusable code using Java packages, interfaces and collection framework.		H											
4. Apply functional programming principles to design server less web services.	H												
5. Design object-oriented solution using UML for real-world problems using system thinking.			H										

Degree of compliance L: Low M: Medium H: High



Competency addressed in the Course and corresponding Performance Indicators

Competency	Performance Indicators	Course Specific Performance Indicators
1.3 Demonstrate competency in engineering fundamentals.	1.3.3 Apply the principles of computer science including algorithms, data structures, and programming to develop a solution for complex engineering problems.	1.3.3.1 Apply OOP principles, java platform features to develop java programs using basic data types. 1.3.3.2 Interpret problem statements and illustrate appropriate Java class structures, method logic, and/or object interactions, methods with objects as parameters and nested classes for solving moderately complex engineering problems.
2.1: Demonstrate an ability to identify and define complex engineering problems by understanding real-world contexts and constraints.	2.1.2: Identify critical systems/sub systems, variables, and constraints that characterize the problem environment by analyzing practical contexts and real-world limitations.	2.1.2.1 Identify and Apply inheritance, method overriding, dynamic method dispatch, abstract class, interfaces and exception concept to implement the given problem. 2.1.2.2 Apply lambda expressions and/or streams API to solve a problem. 2.1.2.3 Apply spring boot/RESTful API to build a solution for real world problems. 2.1.2.4 Apply appropriate collection interfaces and classes (List, Set, Map, and their implementations) to organize, store, and manipulate data efficiently based on problem requirements and usage scenarios.
3.3: Demonstrate an ability to select an optimal design solution considering cost-effectiveness and environmental impact.	3.2.2: Build models/prototypes to develop chosen design solutions.	3.2.2.1 Identify the relationship between the classes/objects to establish associations/aggregation/composition in the design and interpret the design effectively 3.2.2.2 Decompose complex problem scenarios into well-defined classes and subsystems by applying appropriate design patterns (suitable structural and behavioral design patterns) to manage responsibilities and object creation.

Eg: 1.2.3: Represents Program Outcome '1', Competency '2' and Performance Indicators '3'.



Course Content

Course Code: 24ECAC207	Course Title: Object Oriented Programming	
L-T-P : 3-0-0	Credits: 3	Contact hours: 3 hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 50	Tutorial/ Practical Hrs : --	Exam Duration: 3 Hrs

Content	Hrs
Unit – 1	
Chapter No 1. Object Oriented Programming (OOP) Concepts and JAVA Language Fundamentals: Introduction to OOP, Object Oriented Paradigm, Applications of OOP. Features of JAVA, JVM, JDK, Source File Structure. Programming constructs and String Handling. Class Diagrams-UML notations. Types of relations between classes.	04 hrs
Chapter No 2. Classes and Objects Class Fundamentals, Declaring objects, Assigning object reference variables, Introducing methods, Constructors, this key word, Garbage collection: The finalize method. A closer Look at Methods and Classes: Overloading: Methods, Constructors. Using objects as Parameters, Returning objects, Access control. Understanding static and final keywords. Introducing nested and inner classes.	06 hrs.
Chapter No. 3 Inheritance and Polymorphism Inheritance: Basics, Usage of super key word, Method overriding, Dynamic method dispatch, Abstract classes, Object class.	06 hrs.
Unit – 2	
Chapter No. 4 Packages, Interfaces & Exception handling Packages and Interfaces: Packages, Access protection, Importing packages. Interfaces. Exception Handling: Fundamentals, Exception Types, Using try, catch, throw, throws and finally, Multiple catch, Nested try statements, and user defined exceptions.	06 hrs.
Chapter No. 5: Collections Framework: Introduction, List Interface, List Implementation Classes, Set Interface, Set Implementation Classes, The Map Interface, Map Implementation Classes.	04 hrs.
Chapter No. 6: Lambda Expressions & Streams API: Functional programming, Functional interface, Bulk operations on collections, Basics of Streams, Reduction operations, Iterators and Streams	06 hrs.
Unit – 3	
Chapter No. 7 Design Patterns: Creational, Structural and Behavioral design patterns.	4 hrs.
Chapter No. 8: Spring Boot: Overview, key features, architecture, setting-up a Spring Boot, RESTful API, spring data JPA, CRUD Operations.	4 hrs.



Textbooks

1. Herbert Schildt, "The Complete Reference", 9th Edition, McGraw-Hill.

Reference Books

1. Kathy Sierra and Bert Bates, Head First JAVA, 2, O'Reilly Media.
2. Introduction to Java Programming, Liang Y D, Pearson, 7th Edition.

Evaluation Scheme

ISA Scheme

Assessment	Weightage in Marks
ISA-1	20
ISA-2	20
Activity (UML Design)	10
Total	50

Course Unitization for Minor Exams and End Semester Assessment

Topics / Chapters	Teaching Hours	No. of Questions in ISA-1	No. of Questions in ISA-2	No. of Questions in Activity	No. of Questions in ESA
Unit I					
Chapter No 1. Object Oriented Programming (OOP) Concepts and JAVA Language Fundamentals	4	1.00	-	-	1.00
Chapter No 2. Classes and Objects	6	1.00	-	-	1.00
Chapter No. 3 Inheritance and Polymorphism	6	1.00	-	-	1.00
Unit II					
Chapter No. 4 Packages , Interfaces &Exception handling	6	-	1.5	-	1.5
Chapter No. 5: Collections Framework	4	-	0.5	-	0.5
Chapter No. 6: Lambda Expressions & Streams API	6	-	1.00	-	1.00
Unit III					
Chapter No. 7 Design Patterns	4	-	-	1.00	1.00
Chapter No. 8: Spring Boot	4	-	-	-	1.00



Note

1. Each Question carries 20 marks and may consists of sub-questions.
2. Mixing of sub-questions from different chapters within a unit (only for Unit I and Unit II) is allowed in ISA I, II and ESA
3. Answer 5 full questions of 20 marks each (two full questions from Unit I, II and one full questions from Unit III) out of 8 questions in ESA.

Course Assessment Plan

Course Title: Object Oriented Programming Code: 24ECAC207					
Course outcomes (COs)	Weightage in assessment	Assessment Methods			
		ISA 1 Exam	ISA 2 Exam	Activity (Design)	End semester Assessment
1. Describe the object-oriented programming paradigm.	12%	✓			✓
2. Apply object oriented principles to design a solution using JAVA.	25%		✓		✓
3. Design reusable code using Java packages, interfaces and collection framework.	25%		✓		✓
4. Apply functional programming principles to design server less web services.	20%		✓		✓
5. Design object-oriented solution using UML for real-world problems using system thinking.	17%	✓	✓	✓	✓
Weightage		20%	20%	10%	50%



Course Activity Rubrics:

Parameter	Marks	PI	CO	BL
Identify classes and relationships	5	3.2.2	5	L4
Draw detailed Class diagrams	5	3.2.2	5	L4

Parameters	Marks	Excellent(4-5M)	Good(2-3M)	Poor(0-1M)
Design- Identify classes and relationships PI: 3.2.2 CO: 5 BL: L4	5	The student is able to identify all type of classes, relationships and design patterns.	The student is able to identify most type of classes, relations and design pattern.	The student is unable to identify key type of classes, relations and design pattern.
Design- Draw detailed Class diagrams PI: 3.2.2 CO: 5 BL: L4	5	The student is able to draw the complete class diagram using UML notations and design patterns.	The student is able to draw the partial class diagram using UML notations and partial design pattern.	The student is unable to draw the class diagram.

Date:

HOD, CSE(AI)



Chapter wise Plan

Course Title and Code: Object Oriented Programming(24ECAC207)	
Chapter Number and Title: 1. Object Oriented Programming (OOP) Concepts and JAVA Language Fundamentals	Planned Hours:4hrs

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Differentiate between structured and object Oriented Programming.	1	L3	1.3.3
2. Explain the principles of Object Oriented Programming (OOP).	1	L2	1.3.3
3. Explain java platform features.	1	L2	1.3.3
4. Write java programs using arrays, strings and basic types.	1	L3	1.3.3
5. Draw the class diagram using proper UML notations	5	L3	3.3.2

Lesson Schedule
Class No. - Portion covered per hour
1. Introduction to OOP, Object Oriented Paradigm, Applications of OOP.
2. Features of JAVA, JVM, JDK, Source File Structure.
3. Programming constructs, arrays.
4. String Handling
5. Class Diagrams-UML notations.
6. Types of relations between classes.

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. Compare and contrast structured and object oriented programming paradigms.	1	L2	1.3.3
2. List and explain OOP features.	1	L2	1.3.3
3. Explain java features.	1	L2	1.3.3
4. Compare procedural programming vs. OOP from a system boundary perspective.	1	L3	1.3.3
5. Explain arrays? Write a sample code to demonstrate array of basic type and its usage.	3	L3	1.3.3
6. How many objects will be created in the following code? String s1="kletech"; String s2="kletech";	3	L2	1.3.3



7. Explain that the java is platform independent programming language.	2	L2	1.3.3
8. Demonstrate the strings are immutable in java.	3	L3	1.3.3
9. Differentiate string and StringBuffer classes with example.	3	L3	1.3.3
10. Write the output of the following code snippet. <pre>class StringComparisonUsingEqualsMethod{ public static void main(String args[]){ String s1="Sachin"; String s2="Sachin"; String s3=new String("Sachin"); String s4="Saurav"; System.out.println(s1.equals(s2));//true System.out.println(s1.equals(s3));//true System.out.println(s1.equals(s4));//false } }</pre>	3	L3	1.3.3
11. Explain the Object-Oriented Paradigm and discuss how it supports real-world modelling with suitable examples.	1	L2	1.3.3
12. Describe the roles of JVM, JRE, and JDK in the Java execution environment. How do they interact during program execution?	2	L3	1.3.3
13. Write a Java program to perform basic string operations such as length calculation, concatenation, and character extraction.	3	L3	1.3.3
14. Draw a simple UML class diagram for a Student–Course system and explain the type of relationship used between the classes.	5	L3	3.2.2
15. What happens to the entire software system if the heap memory is exhausted?	3	L3	1.3.3



Course Title and Code: **Object Oriented Programming(24ECAC207)**

Chapter Number and Title: **2. Classes and Objects**

Planned Hours: **6hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1.Explain classes and objects.	2	L2	1.3.3
2.Define constructor and overloading of method to solve a given problem.	2	L2	1.3.3
3.Apply static members concept to solve a given problem.	2	L3	1.3.3
4.Define methods with objects as parameter and return value.	2	L2	1.3.3
5.Apply nested and inner classes concepts to solve a given problem.	2	L3	1.3.3
6.Draw class diagrams using UML notations for a given scenario.	2	L3	3.2.2

Lesson Schedule

Class No. - Portion covered per hour

1. Class Fundamentals, Declaring objects, Assigning object reference variables.
2. Introducing methods, Constructors, this key word, Garbage collection: The finalize method.
3. A closer Look at Methods and Classes.
4. Overloading: Methods, Constructors.
5. Overloading: Methods, Constructors.
6. Using objects as Parameters, Returning objects.
7. Access control. Understanding static and final keyword.
8. Nested and inner classes.

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. What is class and object? Write a program demonstrating access of same object with more than one reference?	1	L3	1.3.3
2. Discuss i) Default constructor ii) Parameterized constructor iii) Copy constructor with examples.	1	L2	1.3.3
3. Demonstrate passing array of objects to a function with sample code.	1	L3	1.3.3
4. Create a class Point with members intx,y. Create two Point objects and initialize them different values for x and y. Then swap the member values of the objects with each other.	2	L3	1.3.3
5. How does overloading improve system usability and flexibility?	2	L3	1.3.3



Give example.			
6. What is the difference between a local variable and a data member?	1	L2	1.3.3
7. Explain the need of method overloading with a suitable example.	2	L3	1.3.3
8. A cricket player information needs to be maintained such as name, no of innings, total runs scored and batting average. Write a java program to satisfy following requirements I. Information of at least 5 players needs to maintained II. Player information can be displayed whenever required. III. Batting average of a player can be updated anytime. IV. Player with highest batting average can be displayed form a set of players. V. Players are initialized whenever they are created.	4	L3	1.3.3
9. Consider simple library application. library has some books and librarian can issue the books to students. Library can list the books that are issued to the students with their details. i. Create a Library class with members collegeName, librarian, books. Include methods to issue books to students and to display list of books that are issued to students. ii. Create a class Book with members title, publisher, student-issued to- and display method. iii. Create a class Student with members name, srn, mobileNum and display method. iv. Identify relationship between the classes. Include appropriate parameterized constructors in each class.	4	L3	1.3.3
10. Explain the different types of has-relationships between objects and consider the following objects of banking domain • BankAccount(accNo, accBal, accType) • AadharCard (aadharNo, name, address) • PanCard(panNo, name) The government India announces for its citizens to link the objects correspondingly for smooth transactions of their bank accounts. Perform the following: i. Identify the relationships between the given objects and create the class diagram accordingly, the program should reject if more than one adhaar or pan is linked correspondingly. ii. Include a method to print the account details and the related objects data.	6	L3	3.2.2



11. Explain the significance of the this keyword with an example.	1	L2	1.3.3
12. Explain method overloading and compile-time resolution with a program.	2	L3	1.3.3
13. Write a Java program using static data members and static methods.	3	L3	1.3.3
14. Explain how class diagrams contribute to system scalability.	5	L3	1.3.3
15. Identify classes, relationships, and draw a UML class diagram for a Company–Employee system.	6	L3	3.2.2



Course Title and Code: Object Oriented Programming(24ECAC207)	
Chapter Number and Title: 3. Inheritance and Polymorphism	Planned Hours: 6hrs

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1.Explain inheritance and polymorphism.	2	L2	2.1.2
2.Explain different types of inheritance supported by java.	2	L2	2.1.2
3.Apply method overriding and dynamic method dispatch.	2	L2	2.1.2
4.Identify and create abstract classes to solve a given scenario.	2	L3	2.1.2
5. Draw the class diagram for the given class hierarchy.	2	L3	3.2.2

Lesson Schedule
Class No. - Portion covered per hour
1. Inheritance: Basics, types, application of inheritance and UML
2. Usage of super key word
3. Method overriding
4. Dynamic method dispatch
5. Abstract classes
6. Object class

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. Define inheritance. Explain different types of inheritance supported by java.	1	L2	2.1.2
2. How constructors of parent class are invoked? Illustrate with a program?	1	L3	2.1.2
3. Discuss trade-offs between flexibility and complexity in inheritance hierarchies.	3	L2	2.1.2
4. Consider a class hierarchy Employee(empId, name, salary, bonus) Manager(numberOfShares) and Salesman(numberOfSales). Assume initial bonus amount is zero for all employees. Define a method to pay bonus and consider type of employee while paying bonus. Write a java program to create a structure of all the classes.	3	L3	2.1.2



5. Create a class hierarchy of classes CashCards (bankName, void swipe ()), CreditCard and DebitCard. The credit and debit cards can be used to pay shopping amount and DebitCard can be used to pay balance generated by CreditCard. Identify suitable/missing data members and define appropriate constructors. Write a test program to demonstrate runtime polymorphism dynamic method despatch.	4	L3	2.1.2
6. What inheritance relationships would you establish among the following classes and draw the class diagram for any one domain. Student, Professor, TeachingAssistant, Employee, Secretary, DepartmentChair, Janitor, SeminarSpeaker, Person, Course, Seminar, Lecture, ComputerLab.	5	L3	3.2.2
7. Create a super class Car with members color, name, noOfSeats, maxSpeed, speed, accelerate(), break(). Then create two sub-classes Ferrari, BMW. i) include a constructor in Car, Car(name, color, maxSpeed, noOfSpeed) ii) include constructors in classes Ferrari, BMW which reuses the constructor of CAR class iii) over-ride methods accelerate(), break() in sub-classes iv) write a test program to create a object of Ferrari and BMW. Race both the cars.	4	L3	2.1.2
8. Write the output of the following code snippet. abstract class Vehicle { public int speed() { return 0; } class Car extends Vehicle { public int speed() { return 60; } class RaceCar extends Car { public int speed() { return 150; } ... RaceCar racer = new RaceCar(); RaceCar racer = new Car(); Car car = new RaceCar(); Vehicle vehicle = new RaceCar(); System.out.println(racer.speed() + ", " + car.speed() + ", " + vehicle. speed());	4	L3	2.1.2
9. Explain the role of the super keyword in Java. Illustrate how it is used to invoke parent class methods and constructors with an example.	1	L2	2.1.2
10. Differentiate between method overloading and method overriding. Explain why method overriding is essential for	1	L2	2.1.2



achieving runtime polymorphism.			
11. Write a Java program that demonstrates dynamic method dispatch using a base class reference pointing to different derived class objects.	3	L3	2.1.2
12. For a given real-world scenario of a Shape hierarchy (Shape → Circle, Rectangle, Triangle) identify abstract methods, implement an abstract class, and demonstrate polymorphic behaviour using a test program.	4	L3	2.1.2
13. An online shopping system represents products using an inheritance hierarchy. Product is the base class with attributes productId and price, and a method getFinalPrice(). ElectronicProduct and ClothingProduct inherit from Product. ElectronicProduct adds warrantyPeriod and overrides getFinalPrice() to include tax. ClothingProduct adds size and overrides getFinalPrice() to apply discounts. Base class attributes are protected; subclass attributes are private. Draw a UML class diagram for the above hierarchy showing: Generalization (inheritance), Access specifiers, Method overriding.	5	L3	3.2.2
14. You are designing a software system for a University Management System. Entities include <i>Person</i> , <i>Student</i> , <i>Faculty</i> , <i>TeachingAssistant</i> , and <i>Staff</i> . Propose an inheritance hierarchy. Justify your choice of base and derived classes. Identify attributes and methods that should be inherited and those that should be overridden.	5	L3	2.1.2
15. In a Payment Processing System, you have a base class Payment and subclasses CreditCardPayment, UPIPayment, and NetBankingPayment. Which methods should be declared in the base class? Which methods should be overridden in subclasses? How does inheritance support extensibility when new payment modes are added?	5	L3	2.1.2
16. An e-commerce application initially supports payment through Credit Card and UPI. The system is designed such that the checkout module processes payments without			



depending on specific payment methods.			
Later, new payment options like Wallet and Net Banking need to be added without modifying the existing checkout logic.			
a) Explain how polymorphism supports this requirement. b) Identify the role of interface/abstract class in this design. c) Illustrate how the system achieves plug-and-play architecture.			



UNIT II

Course Title and Code: Object Oriented Programming(24ECAC207)	
Chapter Number and Title: 4: Packages , Interfaces & Exception handling	Planned Hours: 06 hrs

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain packages, Interfaces and Exception handling mechanism of java.	3	L2	2.1.2
2. Write java program to create code library using packages.	3	L2	2.1.2
3. Write a java program to manage run time errors using java exception handling mechanism.	3	L3	2.1.2
4. Write a java program to implement interfaces for a given problem specification.	3	L3	2.1.2
5. Draw the class diagram using appropriate UML notations for the given scenario.	3	L3	3.2.2

Lesson Schedule
Class No. - Portion covered per hour
1. Packages: Define, create, access protection and importing packages.
2. Interfaces: Define, create interface and implement.
3. Implementing polymorphism using interfaces.
4. Differences between interfaces and abstract classes.
5. Multiple inheritance and Run time polymorphism using interfaces.
6. Exceptions: Define, Exception handling fundamentals
7. Classification: Exception Types
8. Examples. Using try, catch keywords of exception handling, multiple catch blocks.
9. Using throw, throws, finally keywords to manage exceptions
10. User defined exceptions.

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. Explain why packages in java?	1	L2	2.1.2
2. Explain access protection supported by packages.	1	L2	2.1.2
3. Explain with an example illustrating how an interface reference variable can be used to access an object of a class that implements the interface.	2	L2	2.1.2



4. List the differences between interface and abstract class.	2	L2	2.1.2
5. Define exception and explain exception types.	2	L2	2.1.2
6. Define a class CurrentDate with suitable data members. Define suitable constructor to create CurrentDate object and accept the values from user to create date object. Identify at least two exception/runtime error conditions and apply user defined exceptions concept. Write a test program to create CurrentDate object and manage errors.	2	L2	2.1.2
7. Simulate loan application. The loans are four wheeler loan and two wheeler loan. The customer can avail either of loans and he is provided choice. The application has to provide functionality to calculate EMI, display the customer details along with his loan details and pay EMI until repayment of loan amount.	2	L3	2.1.2
8. Which one of the following class declarations is a valid declaration of a class that cannot be instantiated? Select the one correct answer. (a) class Ghost { abstract void haunt(); } (b) abstract class Ghost { void haunt(); } (c) abstract class Ghost { void haunt() {};} (d) abstract Ghost { abstract void haunt(); } (e) static class Ghost { abstract haunt(); }	2	L3	2.1.2
9. Consider customer can have multiple accounts like SavingsAccount and ProvidentFundAccount, the customer can withdraw and deposit amount to savings account but not with provident fund account (PF account), however he can transfer amount from saving account to PF account. Identify the need of interface in this scenario and create the classes of the class hierarchy. Write a test program to simulate the above scenario.	2	L3	2.1.2
10. Draw a class diagram using appropriate UML notations for the question 7.	3	L3	2.1.2
11. Given the following type and reference declarations, which assignment is legal? // Type declarations: interface I1 {} interface I2 {} class C1 implements I1 {} class C2 implements I2 {} class C3 extends C1 implements I2 {}	2	L3	2.1.2



<pre>// Reference declarations: C1 obj1; C2 obj2; C3 obj3=null; Select the one correct answer. (a) obj2 = obj1; (b) obj3 = obj1; (c) obj3 = obj2; (d) I1 a = obj2; (e) I1 b = obj3; (f) I2 c = obj1;</pre>			
12. Explain how Java packages help in avoiding name conflicts and improving code reusability. Illustrate with an example involving two packages containing classes with the same name.	1	L2	2.1.2
13. A banking application is divided into modules such as account management, transactions, and loan processing. Multiple teams are working on different modules. The system must also support multiple payment methods and handle runtime failures like insufficient balance or network issues. Answer the following Questions: a. How should packages be designed to support modular development? b. How can interfaces be used to integrate different payment methods? c. What type of exception handling strategy should be used to ensure system reliability? d. Explain how these three concepts together improve system scalability.	4	L3	2.1.2
11. Explain the exception handling flow in Java when multiple catch blocks and a finally block are used. Support your answer with a suitable code snippet.	3	L2	2.1.2
12. Design and implement a Java program that uses the throw and throws keywords to handle invalid input conditions in a typical banking application.	3	L3	2.1.2
13. For a Payment System scenario supporting Cash, Card, and UPI payments, identify suitable interfaces and classes, explain how runtime polymorphism is achieved, and draw	5	L3	3.2.2



the corresponding UML class diagram.



Course Title and Code: **Object Oriented Programming(24ECAC207)**

Chapter Number and Title: **5: Collections Framework**

Planned Hours: **4hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the various interfaces and classes provided by collection framework.	3	L2	2.1.2
2. Explain the advantages of collection framework.	3	L2	2.1.2
3. Identify the appropriate collection class required to implement the given scenario.	3	L3	2.1.2
4. Illustrate the methods defined by different interfaces of collection framework.	3	L3	2.1.2

Lesson Schedule

Class No. - Portion covered per hour

1. Introduction to Collection Framework, Interfaces and classes
2. Exploring List Interface
3. Exploring List, set and Queue interface implementation classes
4. Exploring Map Interface and its implementation class

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. Outline the key advantages of collection framework.	2	L2	2.1.2
2. Explain the List and Set interfaces by highlighting important methods.	1	L2	2.1.2
3. Explain the following methods of Queue interface i) Element ii) Offer(E obj) iii) peek() iv) poll() v) remove()	1	L2	2.1.2
4. Suppose setA is a set that contains the strings "CPU", "RAM", and "SSD", and setB is another set that contains the strings "RAM", "GPU", and "HDD". Answer the following questions assuming all the statements are executed sequentially. a) What are the contents of setA and setB after executing setA.addAll(setB)? b) What are the contents of setA and setB after executing setA.add(setB)? c) What are the contents of setA and setB after executing setA.removeAll(setB)? d) What is the content of setA after executing setA.clear()?	4	L3	2.1.2



5. Write a java program to perform the following: i) Create a class Book(bookId, bookTitle, bookAuthor, price) ii) Create a LinkedList object and add few books to it. iii) Call the methods on linkedList to obtain first book, last book added. iv) Call methods to remove the first and last book added.	4	L3	2.1.2
6. Write a program using the appropriate collection for the following scenario which stores the objects sequentially. A bank maintains a list of BankAccount objects. Each account has details such as account number, account holder name, and balance. The bank should be able to display the list of accounts with zero balance and the list of accounts with non-zero balance.	3	L3	2.1.2
7. A system generates logs continuously and needs: - Fast insertion - Occasional search - Sorting by timestamp Answer the following questions: - Which collection is best suited for log storage? - How would you handle large-scale data efficiently? - Discuss performance considerations in your design.			



Course Title and Code: Object Oriented Programming(24ECAC207)	
Chapter Number and Title: 6: Lambda Expressions & Stream API:	Planned Hours: 6hrs

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain functional interface.	4	L2	2.1.2
2. Explain Lambda expressions and stream interface.	4	L2	2.1.2
3. Identify the need of lambda expression/streams API to solve the given problem and implement using lambda expressions/streams API.	4	L3	2.1.2
4. Implement the generic functional interface for the given scenario.	4	L3	2.1.2
5. Explain the reduction operations and give example.	4	L3	2.1.2

Lesson Schedule
Class No. - Portion covered per hour
1. Functional programming
2. Functional interface and Lambda expressions
3. Bulk operations on collections
4. Basics of Stream and stream operations
5. Reduction operations
6. Iterators and Streams

Review Questions

Sl. No. – Questions	TLOs	BL	PI Code
1. Define lambda expression and streams API. Give one example for each.	1	L2	2.1.2
2. // A sample functional interface // (An interface with a single abstract method (SAM)) interface FunctionalInterface { // An abstract function void abstract Function(int x); } Write two different lambda implementation for the given	3	L3	2.1.2



functional interface			
3. Write the following methods that <i>return a lambda expression</i> performing a specified action: A. PerformOperationisOdd(): The lambda expression must return true if a number is odd or false if it is even. B. PerformOperationisPrime(): The lambda expression must return true if a number is prime or false if it is composite. C. PerformOperationisPalindrome(): The lambda expression must return true if a number is a palindrome or false if it is not.	3	L3	2.1.2
4. Write a Java program to implement a lambda expression to find the sum of two integers.	3	L3	2.1.2
5. Identify which is not applicable to Lambda expressions • A lambda expression is an anonymous function • Lambda expressions can be created without belonging to a class • Lambdas were introduced in Java 8 • The -> operator separates a lambda's parameters from the method body • Lambdas can be used inside of forEach() method	2	L2	2.1.2
5. Explain the stream interface.	2	L2	2.1.2
6. Implement the following reduction operations using streams API a) min() b) max() c) sorted()	5	L3	2.1.2
7. Create an ArrayList with the following members: 7,10,18, 24,17,5. Write a java program to reduce the list values to product of list values using streams API.	5	L3	2.1.2
8. Predict the output of the following code: class StreamDemo { public static void main(String[] args) { // Create a list of Integer values. ArrayList<Integer> numList = new ArrayList<>(); numList.add(12); numList.add(3); numList.add(25); numList.add(8); numList.add(19); numList.add(6);	5	L3	2.1.2



<pre>Stream<Integer> numStream = numList.stream(); Optional<Integer>someVal= numStream.min(Integer::compare); if (someVal.isPresent()) System.out.println("value: " + someVal.get()); Stream<Integer> numStream = numList.stream().sorted(); numStream.forEach(n -> System.out.print(n + " "));}}</pre>			
9. A banking system records transactions (credit/debit). Given: import java.util.*; import java.util.stream.*; <pre>class Transaction { int id; String type; // credit or debit double amount; Transaction(int id, String type, double amount) { this.id = id; this.type = type; this.amount = amount; } public String toString() { return "Transaction ID: " + id + ", Type: " + type + ", Amount: " + amount; } }</pre> Answer the following questions: a) How can you identify fraudulent transactions using stream filters? b) How would you calculate total debit amount using streams? c) What challenges arise when processing transactions in parallel streams?			



10. A system generates large log files containing messages.

```
import java.util.*;
import java.util.stream.*;
class Log {
    int id;
    String level; // INFO, ERROR, WARNING
    String message;
    Log(int id, String level, String message) {
        this.id = id;
        this.level = level;
        this.message = message;
    }
    public String toString() {
        return "Log ID: " + id + ", Level: " + level + ", Message: " +
message;
    }
}
```

Write a java program to answer the following Questions:

- How can streams be used to filter error logs?
- How would you count occurrences of different log levels?
- What are the advantages of using streams over traditional loops here?



UNIT III

Course Title and Code: Object Oriented Programming(24ECAC207)	
Chapter Number and Title: 7: Design Patterns	Planned Hours: 4hrs

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain Design patterns and its types.	5	L2	3.2.2
2. Apply a suitable design pattern to solve a given problem.	5	L3	3.2.2
3. Analyze and compare creational, structural, and behavioral design patterns to justify their selection for a given software design scenario.	5	L3	3.2.2

Lesson Schedule
Class No. - Portion covered per hour
1. Creational design patterns
2. Creational design patterns
3. Structural design patterns.
4. Structural design patterns.
5. Behavioural design patterns.

Review Questions

Sl. No. – Questions	TLOs	BL	PI Code
1. Explain Factory Method design pattern with a suitable example.	1	L2	3.2.2
3. What is a Design Pattern? Classify design patterns and explain any two creational patterns.	1	L2	3.2.2
4. Explain the Strategy design pattern with a real-world scenario.	1	L2	3.2.2
5. Describe Singleton design pattern. Explain how it ensures a single instance in Java.	1	L2	3.2.2
6. Consider an online shopping system where different payment methods (Credit Card, UPI, Net Banking) are supported. Apply a suitable design pattern and write a Java program.	2	L3	3.2.2



7. Design a notification system where users receive updates via SM notification. Apply an appropriate design pattern and implement	2	L3	3.2.2
8. Design a logging framework supporting File, Console, and Database logging. Compare applicable design patterns and implement the best choice.	3	L3	3.2.2



Course Title and Code: **Object Oriented Programming(24ECAC207)**

Chapter Number and Title: **8. Spring Boot**

Planned Hours: **4hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the purpose, key features, and architecture of Spring Boot.	4	2	2.1.2
2. Implement RESTful web services using Spring Boot annotations, handle HTTP requests.	4	3	2.1.2
3. Use Spring Data JPA, and perform CRUD operations.	4	3	2.1.2

Lesson Schedule

Class No. - Portion covered per hour

1. Overview, key features, architecture, setting-up a Spring Boot
2. RESTful API
3. Spring data JPA
4. CRUD Operations.
5. Integration of RESTful APIs with Spring Data JPA and CRUD operations (case study / hands-on example)

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. Explain the overview and key features of Spring Boot.	1	L2	2.1.2
2. Describe the architecture of a Spring Boot application with its main components.	1	L2	2.1.2
3. Explain the steps involved in setting up a Spring Boot application using Spring Initializr.	1	L2	2.1.2
4. Explain the role of application.properties or application.yml in Spring Boot applications.	1	L2	2.1.2
5. A college management system needs to provide student details through a web service. Explain how you would implement a RESTful API using Spring Boot annotations to handle HTTP requests.	2	L3	2.1.2
6. An online bookstore application requires support for adding, updating, viewing, and deleting book details. Demonstrate how GET, POST, PUT, and DELETE methods are handled in a Spring Boot REST API.	2	L3	2.1.2



7. A library management system needs to store and retrieve book information from a database. Explain how Spring Data JPA helps in managing data persistence in this scenario.	3	L3	2.1.2
8. In an employee management system, employee records must be created, updated, retrieved, and deleted. Illustrate how CRUD operations are performed using Spring Data JPA in Spring Boot.	3	L3	2.1.2
9. A hospital management system uses a database to manage patient records. With a suitable example, explain the use of repository interfaces in Spring Data JPA.	3	L3	2.1.2
10. Design a simple product management application that exposes RESTful services and performs CRUD operations using Spring Boot and Spring Data JPA.	3	L3	2.1.2



Model Question Paper for Minor Examination –I (ISA)						
Course Code : 24ECAC207		Course Title: Object Oriented Programming				
Duration :75 Mins						
Max. Marks: 40						
	Note: i. Answer any two full questions ii. All questions carry equal marks					
Q.No	Questions	Marks	CO	BL	PO	PI Code
1.a	List features of object oriented programming and explain.	04	1	L2	1	1.3.3
1.b	Given a code snippet, write the output and write your observations. public class MyClass { long var; public void MyClass(long param) { var = param; } public static void main(String[] args) { MyClass a, b; a = new MyClass(); b = new MyClass(5); } }	06	1	L3	1	1.3.3
1.c	Consider a Employee management system, which creates a class hierarchy Employee(empId, name, salary, bonus) Manager(numberOfShares) and Salesman(numberOfSales). Assume initial bonus amount is zero for all employees. Define a method to pay bonus and consider the corresponding attribute of employee while paying bonus. Identify the need of abstract class in this system. Write a java program to implement the class hierarchy and show that abstract class enforce the polymorphism.	10	2	L3	2	2.1.2
2.a	Explain different type of inheritance to implement a reusable code in java.	04	2	L2	1	2.1.2
2.b	Explain the different types of has-relationships between objects and consider the following objects of banking domain • BankAccount(accNo, accBal, accType)	06	5	L3	3	3.2.2



	- AadharCard (aadharNo, name, address) - PanCard(panNo, name) The government India announces for its citizens to link the objects correspondingly for smooth transactions of their bank accounts. Identify the relationships between the given objects and draw the class diagram accordingly.					
2c	Consider a Flight which goes to intermediate destination (upto 2 stops in between the route) before reaching final destination. Each flight can carry fixed number of passengers and particular weights of goods. We can get display list of all the passengers or passengers to particular destination. Write a Java program to create a flight and add passengers to it. Finally print the details of passengers.	10	2	L3	1	1.3.3
3.a	Explain static members of a class.	04	1	L2	1	1.3.3
3.b	Given a code snippet write the correct output: <pre>public class TestPoly { public static void main(String [] args){ Parent p = new Child(); } } class Parent { public Parent() { super(); System.out.println("instantiate a parent"); } } class Child extends Parent { public Child() { System.out.println("instantiate a child"); super(); } }</pre>	06	2	L3	2	2.1.2
3.c	Consider Network service provider like Reliance who provide services like Broadband and DTH connections. Create a class BroadBand with members customerName, id, planAmount.	10	2	L3	2	2.1.2



	• Create a class DTH with member customerMobileNumber, string channelList[], balance. • Create class NetworkProvider with members BroadBandbbArray[], DTH dArray[], bbCount, dthCount. Include methods like registerNewConnection(BroadBand b), registerNewConnection(DTH d), displayConnetions(inttypeOfConnections) Write a main program to create a Network service provider with few connections of broadband and DTH. Then register new connections and display the list.					
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Model Question Paper for Minor Examination –II (ISA-II)						
Course Code: 24ECAC207			Course Title: Course Title: Object Oriented Programming			
Duration		75 Mins				
Max. Marks		40				
	Note: i. Answer any two full questions ii. All questions carry equal marks					
Q.No	Questions	Marks	CO	BL	PO	PI Code
1.a	Explain the exception handling mechanism of java.	04	3	L2	2	2.1.2
1.b	Write the output of the following code snippet and comment on the output. interface Movable { void move(); } interface Stoppable { void stop(); } class Robot implements Movable, Stoppable { public void move() { System.out.println("Robot is moving"); } public void stop() { System.out.println("Robot has stopped"); } } public class MultipleInterfacesExample { public static void main(String[] args) { Movable robot = new Robot(); robot.move(); robot.stop(); } }	06	3	L3	2	2.1.2
1c	Differentiate List and ArrayList and write a java program to explain how these two are related and implemented separately.	10	3	L3	2	2.1.2



2.a	Write the output of the following code snippet and comment on your output. <pre>public class Test { public static void main(String[] args) { try { System.out.printf("1"); int data = 5 / 0; } catch(ArithmeticException e) { System.out.printf("2"); System.exit(0); } finally { System.out.printf("3"); } System.out.printf("4"); } }</pre>	04	3	L3	14	2.1.2
2.b	What is lambda expression and provide syntax of lambda expression with examples.	06	4	L2	2	2.1.2
2.c	Consider the ArrayList as the source of stream to be created and demonstrate the following: - Find the minimum and maximum value - Display only the odd values from the ArrayList	10	3	L3	2	2.1.2
3.a	List 4 method of stream interface and explain how they are used in java programs.	4	4	L2	2	2.1.2
3.b	Write the output of the following code snippet and write your comment on the output. <pre>class StreamDemo2 { public static void main(String[] args) { // Create a list of Integer values. ArrayList<Integer>myList = new ArrayList<>(); myList.add(7); myList.add(18); } }</pre>	06	3	L3	2	2.1.2



	<pre>myList.add(10); myList.add(24); myList.add(17); myList.add(5); // Two ways to obtain the integer product of the elements // in myList by use of reduce(). Optional<Integer>productObj = myList.stream().reduce((a,b) -> a*b); if(productObj.isPresent()) System.out.println("Product as Optional: " + productObj.get()); int product = myList.stream().reduce(1, (a,b) -> a*b); System.out.println("Product as int: " + product); } }</pre>					
3.c	The event coordinator of workshop “Image processing using PYTHON” of plideas2024 (Annual event) plans to develop a EventRegistrationApp{ eventTitle, date, time, registrationFee} for registration of participants to workshop. The registration app accepts valid entries and then registration is successful. Assist him to identify one invalid entry and consider it as exception condition and write solution using Java Make some registrations successful and print successful registrations. Apply the user defined exception concept wherever applicable.	10	3	L3	2	2.1.2



	Model Question Paper for End Semester Assessment (ESA)					
Course Code: 24ECAC207			Course Title: Object Oriented Programming			
Duration		3 Hrs				
Max. Marks		100				
	Note: i. Answer any five questions selecting 2 questions from UNIT-1 and UNIT-2 and 1 question from UNIT-3 ii. Each question carries equal marks.					
	UNIT I					
Q.No	Questions	Marks	CO	BL	PO	PI Code
1a.	Explain the 4 features of Java.	04	1	L2	1	1.3.3
1.b	public class MyClass { long var; public void MyClass(long param) { var = param; } // (1) public static void main(String[] args) { MyClass a, b; a = new MyClass(); // (2) b = new MyClass(5); // (3) } }	06	1	L3	1	1.3.3
1c	Consider a Flight which goes to intermediate destination (upto 2 stops in between the route) before reaching final destination. Each flight can carry fixed number of passengers and particular weights of goods. We can get display list of all the passengers or passengers to particular destination. Write a Java program to create a flight and add passengers to it. Finally print the details of passengers.	10	2	L3	1	1.3.3
2.a	Explain hierarchical inheritance with general syntax to implement in Java.	04	3	L2	1	2.1.2
2.b	Explain the different types of has-relationships between objects and consider the following objects of banking domain • BankAccount(accNo, accBal, accType) • AadharCard (aadharNo, name, address) • PanCard(panNo, name) The government India announces for its citizens	06	5	L3	3	3.2.2



	to link the objects correspondingly for smooth transactions of their bank accounts. Identify the relationships between the given objects and create the classes accordingly.					
2c	Create a class hierarchy with following details: Car with members color, name, noOfSeats, maxSpeed, speed, accelerate(), break(). Then create two sub-classes Ferrari, BMW. write a test program to create an object of Ferrari and BMW. Race both the cars.	10	3	L3	2	2.1.2
3.a	What is the difference between overriding and Overloading? Give examples.	04	3	L2	2	2.1.2
3.b	Given a code snippet write your observation on output. <pre>public class TestPoly { public static void main(String [] args){ Parent p = new Child(); } } class Parent { public Parent() { super(); System.out.println("instantiate a parent"); } } class Child extends Parent { public Child() { System.out.println("instantiate a child"); super(); } }</pre>	06	3	L3	2	2.1.2
3.c	Consider a Multiplex that has multiple screens. Each screen displays a particular movie. Write a Java program to perform following operations. I. To create Multiplex, Screens and movies. II. To add, delete and display screens. III. To change movies displayed by a screen.	10	2	L3	1	1.3.3



UNIT II						
Q.No	Questions	Marks	CO	BL	PO	PI Code
4.a	Explain the class hierarchy of exception handling in java.	04	3	L2	2	2.1.2
4.b	Write the output of the following code snippet and comment on the output. <pre>interface Movable { void move(); } interface Stoppable { void stop(); } class Robot implements Movable, Stoppable { public void move() { System.out.println("Robot is moving"); } public void stop() { System.out.println("Robot has stopped"); } } public class MultipleInterfacesExample { public static void main(String[] args) { Movable robot = new Robot(); robot.move(); robot.stop(); } }</pre>	06	3	L3	2	2.1.2
4.c	Differentiate List and ArrayList and write a java program to explain how these two are related and implemented separately.	10	3	L3	2	2.1.2



5.a	Write the output of the following code snippet and comment. <pre>import java.io.EOFException; import java.io.IOException; class Test { public static void main(String[] args) { try { System.out.printf("1"); int value = 10 / 0; throw new IOException(); } catch(EOFException e) { System.out.printf("2"); } catch(ArithmeticException e) { System.out.printf("3"); } catch(NullPointerException e) { System.out.printf("4"); } catch(IOException e) { System.out.printf("5"); } catch(Exception e) { System.out.printf("6"); } } }</pre>	04	3	L3	2	2.1.2
5.b	What is lambda expression and provide syntax of lambda expression with examples.	06	4	L2	2	2.1.2
5.c	Consider a class hierarchy Employee(empId, name, salary, bonus) Manager(numberOfShares) and Salesman(numberOfSales). Assume initial	10	3	L3	2	2.1.2



	bonus amount is zero for all employees. Define a method to pay bonus and consider attribute of respective classes while paying bonus. Write a java program to implement run time polymorphism.					
6.a	List 4 method of stream interface and explain how they are used in java programs.	4	4	L2	2	2.1.2
6.b	Write the output of the following code snippet and write your comment on the output. <pre>class StreamDemo2 { public static void main(String[] args) { // Create a list of Integer values. ArrayList<Integer>myList = new ArrayList<>(); myList.add(7); myList.add(18); myList.add(10); myList.add(24); myList.add(17); myList.add(5); // Two ways to obtain the integer product of the elements // in myList by use of reduce(). Optional<Integer>productObj = myList.stream().reduce((a,b) -> a*b); if(productObj.isPresent()) System.out.println("Product as Optional: " + productObj.get()); int product = myList.stream().reduce(1, (a,b) -> a*b); System.out.println("Product as int: " + product); } }</pre>	06	4	L3	2	2.1.2
6.c	Each person owns an account. Person performs debit and credit operation with the account he owns. Also some time he/she can transfer money to friends account. Write program to create few person along with their account, then perform debit, credit and transfer operation. Handle exception while transferring amount which should below Rs50000.	10	3	L3	2	2.1.2



UNIT - III						
Q.No.	Questions	Marks	CO	BL	PO	PI Code
7a	What is design Pattern? List types of design pattern and explain most commonly used pattern.	10	5	L2	3	3.2.2
7b	Describe Singleton design pattern. Explain how it ensures a single instance in Java.	10	5	L3	3	3.2.2
8a	Describe the architecture of a Spring Boot application with its main components.	10	4	L2	2	2.1.2
8b	An online bookstore application requires support for adding, updating, viewing, and deleting book details. Demonstrate how GET, POST, PUT, and DELETE methods are handled in a Spring Boot REST API.	10	4	L3	2	2.1.2